

Database Support AI

Enterprise Investigation Report

Workspace: Evaluation Pilot

Database: Evaluation Shipping Read Only (sql_server) [connection_id=049d476c-3402-4c0b-af4b-4f8726612f63]

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Table of Contents

1. Executive Summary
2. Question
3. AI Status
4. Key Findings
5. Top Evidence
6. Procedure Path
7. Root Cause
8. Confidence Score
9. Fix
10. Tests
11. Proof of Fix
12. Rollback
13. Missing Evidence
14. Stage 1 - Understand the Question
15. AI Reasoning Status
16. Stage 2 - Discover Context
17. Evidence-First Target Discovery
18. Candidate Scoring Trace
19. Write Path Ranking
20. Facts, Inferences, and Hypotheses
21. Stage 3 - Generate Investigation Hypotheses
22. Stage 4 - Plan Investigation
23. Stage 5 - Collect Evidence
24. Evidence Gate
25. Suggested Verification Checks
26. Evidence Verification Results
27. Stage 6 - Reason
28. Stage 7 - Generate Dynamic Report
29. Self Validation
30. Detected Intent
31. Extracted Entities
32. Question Understanding
33. Generated Investigation Hypotheses
34. Relevant Object Ranking
35. Investigation Scope
36. Stored Procedure Intelligence
37. Evidence Collected

- 38. Confirmed Missing Related Record Candidates
- 39. Evidence Correlation
- 40. Diagnostic Hypothesis Evaluation
- 41. Why It Happened
- 42. Business Process Graph
- 43. Missing Data Analysis
- 44. Recommended Investigation SQL
- 45. Risks
- 46. References Used
- 47. Missing Information / Clarifying Questions

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Executive Summary

Investigation generated dynamically from detected intent, extracted entities, ranked database objects, stored procedure analysis, retrieved documents, approved knowledge, and safe SQL evidence.

Question

Delivery is complete for shipment SHP-5001; why is the empty-return work order missing?

AI Status

AI Reasoning Summary

Item	Value
AI Reasoning	Disabled
Evidence Package	Not applicable
LLM Claims	0 generated, 0 verified, 0 rejected
Trace	Trace disabled or not available.

Key Findings

Key Findings

Finding	Value
Affected object	eval.transport_work_orders
Business key	Not determined
Current status / duplicate / missing condition	Verify upstream entity and current transition status: 1 row(s) returned.
Parent/supporting object	eval.empty_return_instructions, eval.shipment_milestones, eval.shipments

Top Evidence

Strongest evidence items only. Detailed SQL checks, result samples, and verification SQL are in the full audit report.

Strongest Evidence

Evidence	Type	Source	Summary
Verify upstream entity and current transition status	SQL	Read-only evidence query	1 row(s) returned
Check downstream records under expected or alternate identifiers	SQL	Read-only evidence query	1 row(s) returned
Confirmed Missing Related Record Candidates	SQL	Read-only evidence query	1 row(s) returned
Missing Related Record Summary by Issue Type	SQL	Read-only evidence query	1 row(s) returned

Procedure Path

- ✓ Related procedure only: eval.fn_shipping_active_status
- ✓ Writes affected object: No
- ✓ Evidence: Procedure name appears in collected SQL/correlated evidence.

Root Cause

- ✓ **MISSING_RELATED_RECORD**: expected child or related records are missing for parent records based on discovered metadata relationships.
- ✓ **parent_not_eligible** or **parent_status_not_ready**: evaluate returned parent status/state columns and documented eligibility rules.
- ✓ **procedure_failed** or **batch_failed**: only supported if error-log, job-history, or procedure evidence references the affected relationship.

Confidence Score

- ✓ Overall confidence: 83%
- ✓ + 5 evidence query result(s) returned rows.
- ✓ + 4 read-only query result(s) returned no rows and ruled out alternatives.
- ✓ + Affected object identified from question and evidence: eval.transport_work_orders.
- ✓ + Stored procedure definition evidence confirms relevant calculation logic.

Fix

- ✓ Run the recommended SELECT/EXPLAIN queries and verify the evidence with a DBA. Evidence: SQL evidence plan.
- ✓ Apply the smallest safe fix after confirming the exact broken status, key, procedure, or plan. Evidence: confirmed facts and ranked hypotheses.
- ✓ Do not run write or DDL commands without change approval, rollback plan, and backup validation. Evidence: safety policy.

Tests

Recommended Tests

Test ID	Scenario	Steps	Expected Result	Actual Result	Status
TC-001	Evidence validation	Run recommended SQL	Evidence matches report	Pending	Pending
TC-002	Fix validation	Apply approved fix in lower environment and rerun proof SQL	Issue no longer reproduces	Pending	Pending

Proof of Fix

- ✓ Verify the affected parent record still exists and has the expected business/status state.
- ✓ Verify the expected related child record now exists for the parent using the same relationship columns.
- ✓ Verify the missing-related-record proof query returns zero rows for the affected key or scope.
- ✓ Verify no new relevant error-log or job-history failure rows were created after the fix, when those logs are available.

Rollback

- ✓ Capture before-state rows.

- ✓ Apply fix through versioned script or approved deployment.
- ✓ If validation fails, restore previous procedure/config/index state using the rollback script.
- ✓ Re-run proof SQL and attach output.

Missing Evidence

- ✓ Prove requested entity exists in eval.transport_work_orders: no rows returned
- ✓ Prove reported condition on eval.transport_work_orders.TransportWorkOrdersId: no rows returned
- ✓ Prove reported condition on eval.transport_work_orders.IsActive: no rows returned
- ✓ Prove reported condition on eval.transport_work_orders.ShipmentsId: no rows returned

Stage 1 - Understand the Question

- ✓ Investigation Mode: INVESTIGATION
- ✓ Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning.
- ✓ Investigation Goal: MISSING_DATA
- ✓ User Goal: Answer the user question with database evidence: Delivery is complete for shipment SHP-5001; why is the empty-return work order missing?
- ✓ Working Hypothesis: A downstream record should have been generated but is missing.

AI Reasoning Status

AI-assisted Reasoning

Item	Value
AI-assisted reasoning	Disabled
Reason	AI_REASONING_ENABLED=false
Evidence package sent	No
LLM evidence validation	Not applicable
Evidence citations	Not applicable
PII masking	Ready
PII masking scope	Names, emails, phone numbers, insurance/account identifiers are masked before any LLM evidence package is sent.

Stage 2 - Discover Context

Discovered Objects

Object Type	Name	Columns	Indexes
Table	eval.transport_work_orders	TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_transp
Table	eval.empty_return_instructions	EmptyReturnInstructionsId, ContainerAssignmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_empt
Table	eval.shipment_milestones	ShipmentMilestonesId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_shipm
Table	eval.shipments	ShipmentsId, BookingsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_shipm

Object Type	Name	Columns	Indexes
Table	eval.repair_orders	RepairOrdersId, DamageReportsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_repair
Table	eval.truck_movements	TruckMovementsId, TransportWorkOrdersId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_truck
Procedure	eval.fn_shipping_active_status		

Procedure Analysis

Procedure	Definition	Tables Read	Tables Written	Joins	INSERT	UPDATE	DELETE	MERGE	Loops	Transactions	TRY/CATCH
eval.fn_shipping_active_status	Yes			0	0	0	0	0	0	0	No

Evidence-First Target Discovery

- ✓ Affected Object: eval.transport_work_orders
- ✓ Affected Object Reason: Selected from main problem phrase 'work order missing' (missing); parent and secondary-cause terms do not override the target.
- ✓ Inferred Business Key: Not determined
- ✓ Business Key Reason: No non-primary natural key column was identified.

Candidate Scoring Trace

Selected and Rejected Candidates

Object Type	Name	Score	Decision	Reason
metadata_context	049d476c-3402-4c0b-af4b-4f8726612f63		active	workspace_id=e22377ba-277c-4ea8-b132-4b1566ac4fc5; connection_id=049d476c-3402-4c0b-af4b-4f8726612f63
table	eval.audit_history	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates
table	eval.batch_runs	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates
table	eval.bills_of_lading	4.0	rejected	column name relevance=3; relationship relevance=1; outside top scored metadata candidates
table	eval.bookings	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.carrier_assignments	5.5	rejected	column name relevance=4.5; relationship relevance=1
table	eval.container_assignments	4.0	rejected	column name relevance=3; relationship relevance=1; outside top scored metadata candidates
table	eval.container_events	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.container_master	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates
table	eval.customers	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates
table	eval.customs_holds	4.0	rejected	column name relevance=3; relationship relevance=1; outside top scored metadata candidates
table	eval.customs_releases	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.damage_reports	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.dangerous_goods	4.0	rejected	column name relevance=3; relationship relevance=1; outside top scored metadata candidates
table	eval.demurrage_detention	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.depots	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.empty_return_instructions	13.5	selected	table name relevance=8; column name relevance=4.5; relationship relevance=1
table	eval.equipment_interchange	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.evaluation_marker	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates

Object Type	Name	Score	Decision	Reason
table	eval.exceptions	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates
table	eval.gate_transactions	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.integration_messages	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates
table	eval.invoices	4.0	rejected	column name relevance=3; relationship relevance=1; outside top scored metadata candidates
table	eval.ports	1.5	rejected	column name relevance=1.5; outside top scored metadata candidates
table	eval.rail_movements	5.5	rejected	column name relevance=4.5; relationship relevance=1
table	eval.reefer_readings	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.reefer_settings	2.5	rejected	column name relevance=1.5; relationship relevance=1; outside top scored metadata candidates
table	eval.repair_orders	8.0	selected	table name relevance=4; column name relevance=3; relationship relevance=1
table	eval.shipment_milestones	9.5	selected	table name relevance=4; column name relevance=4.5; relationship relevance=1
table	eval.shipments	8.0	selected	table name relevance=4; column name relevance=3; relationship relevance=1

Write Path Ranking

Ranked Procedures

Rank	Procedure	Score	Writes Affected Object	Reads Affected Object	Relationship	Evidence
1	eval.fn_shipping_active_status	3.0	No	No	none confirmed	Procedure name appears in collected SQL/correlated

Facts, Inferences, and Hypotheses

Reasoning Separation

Type	Finding
Confirmed Fact	Affected object: eval.transport_work_orders.
Confirmed Fact	Verify upstream entity and current transition status: 1 row(s) returned.
Confirmed Fact	Check downstream records under expected or alternate identifiers: 1 row(s) returned.
Confirmed Fact	Confirmed Missing Related Record Candidates: 1 row(s) returned.
Confirmed Fact	Missing Related Record Summary by Issue Type: 1 row(s) returned.
Confirmed Fact	Inspect calculation logic in eval.fn_shipping_active_status: 1 row(s) returned.
Inferred Finding	No procedure was confirmed to directly modify the affected object; procedure-related causes remain lower confidence.
Inferred Finding	At least one read-only evidence query returned rows, so reasoning should prioritize observed data over metadata-only matches.
Hypothesis	Additional evidence is required before a root cause can be confirmed.

Stage 3 - Generate Investigation Hypotheses

Hypotheses

Hypothesis	Description
H1	Relationship path eval.shipments -> eval.transport_work_orders does not contain the expected related row or transition evidence.
H2	State or status values in eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones may be preventing the expected workflow transition.

Hypothesis	Description
H3	Procedure write/read path may explain the behavior; investigate procedures that directly modify the affected object based on write-path ranking.
H4	Uploaded documents or approved knowledge may describe a business rule or known incident that changes the interpretation of database evidence.

Stage 4 - Plan Investigation

Investigation Plan

Hypothesis	Objects To Inspect	Evidence Required
H1	eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones, eval.shipments, eval.repair_orders	Parent-to-child relationship evidence; R
H2	eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones, eval.shipments, eval.repair_orders	Source row status/state; Procedure bran
H3	eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones, eval.shipments, eval.repair_orders	Stored procedure read/write behavior; P
H4	eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones, eval.shipments, eval.repair_orders	Approved knowledge and uploaded doc

Stage 5 - Collect Evidence

Verify upstream entity and current transition status

ShipmentsId	BusinessKey	Status
2	SHP-5001	Delivered

Check downstream records under expected or alternate identifiers

TransportWorkOrdersId	ShipmentsId	BusinessKey	Status	CorrelationId
3	2	WO-5003	Active	SHIP-WF-3

Confirmed Missing Related Record Candidates

parent_reference	parent_status	child_reference	issue_type
SHP-5001	Delivered	None	MISSING_RELATED_RECORD

Missing Related Record Summary by Issue Type

issue_type	issue_count	example_parent
MISSING_RELATED_RECORD	1	SHP-5001

Prove requested entity exists in eval.transport_work_orders

SQL
SELECT TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive FROM eval.transport_work_orders WHERE (CAST(BusinessKey AS NVARCHAR(100)) = 'SHP-5001')

Prove reported condition on eval.transport_work_orders.TransportWorkOrdersId

SQL
SELECT TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE ((CAST(BusinessKey AS NVARCHAR(100)) = 'SHP-5001') AND (TransportWorkOrdersId = 3))

Prove reported condition on eval.transport_work_orders.IsActive

SQL
SELECT IsActive, TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE ((CAST(BusinessKey AS NVARCHAR(100)) = 'SHP-5001') AND (IsActive = 1))

Prove reported condition on eval.transport_work_orders.ShipmentsId

SQL

```
SELECT ShipmentsId, TransportWorkOrdersId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE ((CAST(BusinessKey AS NVARCHAR(40)) = @BusinessKey) AND (Status = @Status))
```

Inspect calculation logic in eval.fn_shipping_active_status

procedure_name	definition_excerpt
eval.fn_shipping_active_status	CREATE FUNCTION eval.fn_shipping_active_status(@Status NVARCHAR(40)) RETURNS BIT AS BEGIN RETURN CASE WHEN @Status IN ('A', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z') THEN 1 ELSE 0 END

Evidence Gate

Root-cause analysis is allowed only when the supplied key, affected rows, reported condition, and required relationships are supported by connected database evidence.

- ✓ Supplied business key found in returned evidence: SHP-5001.
- ✓ Affected rows were returned by safe SQL evidence.
- ✓ Parent-child relationship was confirmed by metadata or join evidence.
- ✓ Reported condition was reproduced by returned evidence.

Gate Checks

Check	Result
Gate Required	Yes
Issue Reproduced	Yes
Supplied Business Key Exists	Yes
Reported Condition Exists	Yes
Affected Rows Exist	Yes
Parent-Child Relationship Exists	Yes

Suggested Verification Checks

These checks are suggestions only. A user must approve execution. The app validates every SQL statement before running it and allows only SELECT, SHOW, DESCRIBE, DESC, or EXPLAIN.

Suggested Verification Checks

Claim to verify	Purpose
eval.transport_work_orders is the affected object.	Verifies an investigation claim with human-approved read-only SQL.
shipments is the parent/supporting object for eval.transport_work_orders.	Confirms that the supplied business key is being investigated through the discovered parent-child relationship.
Reported condition passes the evidence gate.	Confirms that root-cause analysis is based on a reported condition reproduced from live evidence.

Claim to verify	Purpose
Missing related record condition is reproduced by live database evidence.	Confirms that the supplied business key is being investigated through the discovered parent-child relation
Exact execution path is supported by live operational evidence.	Verifies an investigation claim with human-approved read-only SQL.
Recommended fix is consistent with collected evidence.	Confirms whether the selected stored procedure is supported by procedure metadata for this claim.
Proof and investigation SQL are valid read-only statements.	Confirms that generated proof and investigation SQL remains read-only.

Evidence Verification Results

The verification agent checks root-cause claims, recommendations, and hypotheses against collected metadata, procedure analysis, documents, and read-only live SQL. It never executes writes or stored procedures.

Evidence Verification Results

Claim	Purpose	Evidence logic	SQL executed	Expected result	Actual result summary	Interpretation	Conclusion	Status
No checks have been executed yet								Pending

Stage 6 - Reason

Ranked Hypotheses

Rank	Hypothesis	Description
1	H1	Relationship path eval.shipments -> eval.transport_work_orders does not contain the expected related row or transition evidence.
2	H2	State or status values in eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones may be preventing the expected workflow tra
3	H3	Procedure write/read path may explain the behavior; investigate procedures that directly modify the affected object based on write-path ranking.
4	H4	Uploaded documents or approved knowledge may describe a business rule or known incident that changes the interpretation of database evidence.

Stage 7 - Generate Dynamic Report

✓ This report was generated after context discovery, hypothesis generation, evidence planning, evidence collection, reasoning, and self-validation.

Self Validation

Validation Checks

Question	Answer
What is the main problem phrase?	Captured in Evidence-First Target Discovery reason.

Question	Answer
What object is missing/duplicated/slow/failed?	eval.transport_work_orders.
Which terms are only possible causes?	Listed in target discovery when secondary cause markers are present.
Did I accidentally choose a secondary cause as the target?	No if the affected object comes from the main problem phrase; review target reason.
Does the highest-ranked hypothesis explain observed evidence?	Yes
Is another hypothesis stronger?	Possible; no direct write path was confirmed
Was the procedure that directly modifies eval.transport_work_orders investigated?	No direct modifier confirmed; continue investigation
Does the chosen root cause write the affected table?	No
Does the error log support the chosen root cause?	No; confidence remains lower until error-log evidence is found
Does the business process graph include the chosen procedure?	No
Does procedure analysis match the report narrative?	No direct write narrative allowed

Detected Intent

- ✓ Intent: MISSING_DATA
- ✓ Confidence: 75%
- ✓ Rationale: Intent inferred from the primary request first, with secondary listed causes weighted lower.

Extracted Entities

Entities

Type	Value
exact_id_or_code	SHP-5001
business_identifier	SHP-5001
possible_table_or_column	Delivery
possible_table_or_column	is
possible_table_or_column	complete
possible_table_or_column	for
possible_table_or_column	shipment
possible_column	SHP
possible_table_or_column	why
possible_table_or_column	the
possible_table_or_column	empty
possible_table_or_column	return
possible_table_or_column	work
possible_table_or_column	order

Question Understanding

- ✓ User Goal: Answer the user question with database evidence: Delivery is complete for shipment SHP-5001; why is the empty-return work order missing?
- ✓ User Hypothesis: A downstream record should have been generated but is missing.
- ✓ Business Process: delivery
- ✓ Likely Objects: eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones, eval.shipments, eval.repair_orders, eval.truck_movements
- ✓ Required Evidence: Rows for the affected business key(s); Relevant table relationships and indexes; Stored procedure read/write behavior; Job, audit, or error-log evidence when available; Approved knowledge and uploaded documents when relevant

Generated Investigation Hypotheses

Hypotheses

Hypothesis	Description
H1	Relationship path eval.shipments -> eval.transport_work_orders does not contain the expected related row or transition evidence.
H2	State or status values in eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones may be preventing the expected workflow transition.
H3	Procedure write/read path may explain the behavior; investigate procedures that directly modify the affected object based on write-path ranking.
H4	Uploaded documents or approved knowledge may describe a business rule or known incident that changes the interpretation of database evidence.

Relevant Object Ranking

Ranked Objects

Object Type	Name	Score	Reason
table	eval.transport_work_orders	31.95	Matched 3 question/entity token(s); selected from main problem phrase target; has foreign-key relationships; has indexes
table	eval.empty_return_instructions	21.45	Matched 2 question/entity token(s); selected from main problem phrase parent; has foreign-key relationships; has indexes
table	eval.shipment_milestones	12.45	Matched 1 question/entity token(s); has foreign-key relationships; has indexes
table	eval.shipments	10.95	Matched 1 question/entity token(s); has foreign-key relationships; has indexes
table	eval.repair_orders	10.95	Matched 1 question/entity token(s); has foreign-key relationships; has indexes
table	eval.truck_movements	9.45	Matched 2 question/entity token(s); has foreign-key relationships; has indexes
procedure	eval.usp_shipping_workflow_1	1.0	Procedure name matched selected object or question tokens
procedure	eval.usp_shipping_workflow_2	1.0	Procedure name matched selected object or question tokens
procedure	eval.usp_shipping_workflow_3	1.0	Procedure name matched selected object or question tokens
procedure	eval.usp_shipping_workflow_4	1.0	Procedure name matched selected object or question tokens
procedure	eval.usp_shipping_workflow_5	1.0	Procedure name matched selected object or question tokens
procedure	eval.usp_shipping_workflow_6	1.0	Procedure name matched selected object or question tokens
procedure	eval.usp_shipping_workflow_7	1.0	Procedure name matched selected object or question tokens
procedure	eval.usp_shipping_workflow_8	1.0	Procedure name matched selected object or question tokens

Investigation Scope

Discovered Objects

Object Type	Name	Columns	Indexes
Table	eval.transport_work_orders	TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_transp
Table	eval.empty_return_instructions	EmptyReturnInstructionsId, ContainerAssignmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_empt
Table	eval.shipment_milestones	ShipmentMilestonesId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_shipm
Table	eval.shipments	ShipmentsId, BookingsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_shipm
Table	eval.repair_orders	RepairOrdersId, DamageReportsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_repar
Table	eval.truck_movements	TruckMovementsId, TransportWorkOrdersId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive	IX_truck
Procedure	eval.fn_shipping_active_status		

Stored Procedure Intelligence

Procedure Analysis

Procedure	Definition	Tables Read	Tables Written	Joins	INSERT	UPDATE	DELETE	MERGE	Loops	Transactions	TRY/CATCH
eval.fn_shipping_active_status	Yes			0	0	0	0	0	0	0	No

Evidence Collected

Verify upstream entity and current transition status

ShipmentsId	BusinessKey	Status
2	SHP-5001	Delivered

Check downstream records under expected or alternate identifiers

TransportWorkOrdersId	ShipmentsId	BusinessKey	Status	CorrelationId
3	2	WO-5003	Active	SHIP-WF-3

Confirmed Missing Related Record Candidates

parent_reference	parent_status	child_reference	issue_type
SHP-5001	Delivered	None	MISSING_RELATED_RECORD

Missing Related Record Summary by Issue Type

issue_type	issue_count	example_parent
MISSING_RELATED_RECORD	1	SHP-5001

Prove requested entity exists in eval.transport_work_orders

SQL

```
SELECT TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive FROM eval.transport_work_orders WHERE (CAST(BusinessKey AS NVARCHAR(50)) = 'SHP-5001')
```

Prove reported condition on eval.transport_work_orders.TransportWorkOrdersId

SQL

```
SELECT TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE ((CAST(BusinessKey AS NVARCHAR(50)) = 'SHP-5001') AND (TransportWorkOrdersId = 3))
```

Prove reported condition on eval.transport_work_orders.IsActive

SELECT IsActive, TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE ((CAST(BusinessKey AS

SQL

```
SELECT ShipmentsId, TransportWorkOrdersId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE ((CAST(BusinessKey AS NVARCHAR(100)) = @BusinessKey) AND (@Status IS NULL OR Status = @Status) AND (@EventTime IS NULL OR EventTime <= @EventTime) AND (@Details IS NULL OR Details = @Details) AND (@CorrelationId IS NULL OR CorrelationId = @CorrelationId))
```

procedure_name	definition_excerpt
----------------	--------------------

eval.fn_shipping_active_status	CREATE FUNCTION eval.fn_shipping_active_status(@Status NVARCHAR(40)) RETURNS BIT AS BEGIN RETURN CASE WHEN @Status IN ('A
--------------------------------	---

Missing Related Record Candidates

parent_reference	parent_status	child_reference	issue_type
SHP-5001	Delivered	None	MISSING_RELATED_RECORD

issue_type	issue_count	example_parent
------------	-------------	----------------

issue_type	issue_count	example_parent
MISSING_RELATED_RECORD	1	SHP-5001

Correlated Evidence

Type	Subject	Finding	Support
SQL	Verify upstream entity and current transition status	1 row(s) returned	ShipmentsId
SQL	Check downstream records under expected or alternate identifiers	1 row(s) returned	TransportWorkOrdersId
SQL	Confirmed Missing Related Record Candidates	1 row(s) returned	parent_reference
SQL	Missing Related Record Summary by Issue Type	1 row(s) returned	issue_type=
SQL	Prove requested entity exists in eval.transport_work_orders	No rows returned	SELECT TransportWorkOrdersId
SQL	Prove reported condition on eval.transport_work_orders.TransportWorkOrdersId	No rows returned	SELECT TransportWorkOrdersId
SQL	Prove reported condition on eval.transport_work_orders.IsActive	No rows returned	SELECT IsActive
SQL	Prove reported condition on eval.transport_work_orders.ShipmentsId	No rows returned	SELECT ShipmentsId
SQL	Inspect calculation logic in eval.fn_shipping_active_status	1 row(s) returned	procedure_name
Procedure	eval.fn_shipping_active_status	Reads 0 table(s), writes 0 table(s), complexity Low, locking risk Low	Reads: none

Ranked Hypotheses

Rank	Hypothesis	Description
1	H1	Relationship path eval.shipments -> eval.transport_work_orders does not contain the expected related row or transition evidence.
2	H2	State or status values in eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones may be preventing the expected workflow tra
3	H3	Procedure write/read path may explain the behavior; investigate procedures that directly modify the affected object based on write-path ranking.
4	H4	Uploaded documents or approved knowledge may describe a business rule or known incident that changes the interpretation of database evidence.

Why It Happened

- ✓ User-reported condition was mapped to relevant database objects.
- ✓ Read-only evidence was collected from ranked tables, procedures, logs, documents, or knowledge.
- ✓ No procedure was confirmed to write eval.transport_work_orders; root-cause wording must stay at likely/unknown confidence.
- ✓ Most likely explanation: Relationship path eval.shipments -> eval.transport_work_orders does not contain the expected related row or transition evidence.
- ✓ Conclusion remains bounded by the supporting and missing evidence listed for the hypothesis.

Business Process Graph

Execution-order graph inferred from stored procedure read/write metadata only.

Inferred Execution Graph

From	To
Unable to infer	Additional procedure read/write evidence required

Missing Data Analysis

- ✓ Identify the expected record, upstream dependency, guard condition, and validation SQL.

Recommended Investigation SQL

Verify upstream entity and current transition status

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

```
SELECT ShipmentsId, BusinessKey, Status FROM eval.shipments WHERE BusinessKey = 'SHP-5001'
```

Check downstream records under expected or alternate identifiers

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

```
SELECT c.TransportWorkOrdersId, c.ShipmentsId, c.BusinessKey, c.Status, c.CorrelationId FROM eval.shipments p JOIN eval.transport_work_o
```

Confirmed Missing Related Record Candidates

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

SELECT

```
    p.BusinessKey AS parent_reference,  
    p.Status AS parent_status,  
    NULL AS child_reference,  
    CASE  
        WHEN c.TransportWorkOrdersId IS NULL THEN 'MISSING_RELATED_RECORD'  
        ELSE 'OK'  
    END AS issue_type  
FROM eval.shipments p  
LEFT JOIN eval.transport_work_orders c ON c.ShipmentsId = p.ShipmentsId AND (LOWER(CAST(c.BusinessKey AS NVARCHAR(MAX)))) LIKE '%work%' C  
WHERE c.TransportWorkOrdersId IS NULL AND p.BusinessKey = 'SHP-5001'  
ORDER BY p.BusinessKey
```

Missing Related Record Summary by Issue Type

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

SELECT

```
    issue_type,  
    COUNT(*) AS issue_count,  
    MIN(parent_reference) AS example_parent  
FROM (  
    SELECT  
        p.BusinessKey AS parent_reference,  
        CASE  
            WHEN c.TransportWorkOrdersId IS NULL THEN 'MISSING_RELATED_RECORD'  
            ELSE 'OK'  
        END AS issue_type  
    FROM eval.shipments p  
    LEFT JOIN eval.transport_work_orders c ON c.ShipmentsId = p.ShipmentsId AND (LOWER(CAST(c.BusinessKey AS NVARCHAR(MAX)))) LIKE '%work%' C  
    WHERE c.TransportWorkOrdersId IS NULL AND p.BusinessKey = 'SHP-5001'  
) missing_related_candidates  
GROUP BY issue_type  
ORDER BY issue_count DESC
```

Prove requested entity exists in eval.transport_work_orders

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

SELECT TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId, IsActive FROM eval.transport_work_orders

Prove reported condition on eval.transport_work_orders.TransportWorkOrdersId

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

SELECT TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE

Prove reported condition on eval.transport_work_orders.IsActive

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

SELECT IsActive, TransportWorkOrdersId, ShipmentsId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders

Prove reported condition on eval.transport_work_orders.ShipmentsId

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

SELECT ShipmentsId, TransportWorkOrdersId, BusinessKey, Status, EventTime, Details, CorrelationId FROM eval.transport_work_orders WHERE

Inspect calculation logic in eval.fn_shipping_active_status

Expected Result: Rows should confirm or rule out the suspected issue. If no rows return, evidence is missing or object names differ.

Risk: Read-only

Risks

- ✓ Business impact depends on the affected process and returned evidence.
- ✓ Technical risk increases if manual data repair is attempted without dependency checks.

References Used

- ✓ Tables: eval.transport_work_orders, eval.empty_return_instructions, eval.shipment_milestones, eval.shipments, eval.repair_orders, eval.truck_movements
- ✓ Procedures: eval.fn_shipping_active_status
- ✓ Views: None discovered
- ✓ Documents: No uploaded documents found

Missing Information / Clarifying Questions

- ✓ Prove requested entity exists in eval.transport_work_orders: no rows returned
- ✓ Prove reported condition on eval.transport_work_orders.TransportWorkOrdersId: no rows returned
- ✓ Prove reported condition on eval.transport_work_orders.IsActive: no rows returned
- ✓ Prove reported condition on eval.transport_work_orders.ShipmentsId: no rows returned